AGA KHAN UNIVERSITY EXAMINATION BOARD

SECONDARY SCHOOL CERTIFICATE

CLASS X EXAMINATION

MAY 2015

Computer Science Paper II

Time: 2 hours 25 minutes Marks: 40

INSTRUCTIONS

Please read the following instructions carefully.

1. Check your name and school information. Sign if it is accurate.

I agree that this is my name and school. **Candidate's signature**

2. RUBRIC. There are TEN questions. Answer ALL questions. Questions 9 and 10 each offer TWO choices. Attempt any ONE choice from each.

3. When answering the questions:

   Read each question carefully.
   Use a black pencil for diagrams. DO NOT use coloured pencils.
   DO NOT use staples, paper clips, glue, correcting fluid or ink erasers.
   Complete your answer in the allocated space only. DO NOT write outside the answer box.

4. The marks for the questions are shown in brackets ( ).

Q.1. (Total 3 Marks)

If A = 0 and B = 1, prove the following Boolean algebra laws.

i. $\overline{(A + B)} = \overline{A}.\overline{B}$ (1 Mark)

ii. $\overline{\overline{(A.B)}} = A.B$ (2 Marks)

Q.2. (Total 4 Marks)

a. Write TWO reasons for using the DIM statement in one and two dimensional array. (2 Marks)

b. There are syntax errors in the following arrays declaration. (2 Marks)

10 N$(6, 7) A(-6)

Rewrite the correct syntax.

Q.3. (Total 3 Marks)

Write GW BASIC statements to print a circle having point of centre (50, 100) with radius 40 in high resolution mode.

Q.4. (Total 5 Marks)

a. Write ONE similarity and ONE difference between FOR NEXT and WHILE WEND statements. (2 Marks)

b. Consider the following GW BASIC program. (3 Marks)

```
10 CLS
20 A = 2
30 PRINT A
40 A = A + 2
50 ????
60 END
```

Write a suitable conditional statement at line 50 to print even numbers up to 100.

Q.5. (Total 3 Marks)

If X = 45 and Y = 67, write a pseudo code to interchange or swap the values of X and Y.

Q.6. (Total 2 Marks)

Write any TWO examples of syntax error in GW BASIC.

Q.7. (Total 3 Marks)

What is the purpose of logical operators in programming? Give TWO examples of logical operators which are used in GW BASIC programming.

Q.8. (Total 2 Marks)

Convert the following mathematical expression into a GW BASIC statement.

$C = \sqrt{a^2 + b^2}$

Q.9. (Total 7 Marks)

EITHER

a. The Body Mass Index (BMI) is a simple weight-for-height (kg/m²) index. It is an important screening tool that tells whether you are obese, overweight, underweight or healthy.

BMI (kg/m^2) Asian	Weight Category	Health Risk
27.5 and above	Obese	Health risk of developing heart disease, high blood pressure, stroke and diabetes
23 to 27.4	Overweight	Moderate risk of developing the above diseases
18.5 to 22.9	Healthy	Low risk of developing the above diseases
Below 18.5	Underweight	At risk of developing nutritional deficiency diseases and osteoporosis

Write a GW BASIC program to accept weight in kg (W) and height in meters (H) of a person as input. Also calculate the BMI using the given formula and print appropriate message on the given criteria in the following table.

$$BMI = \frac{W}{H^2}$$

BMI (kg/m^2)	Weight Category
>= 27.5	Obese
>= 23 AND < = 27.4	Overweight
>= 18.5 AND < = 22.9	Healthy
< 18.5	Underweight

OR

b. Write a GW BASIC program to take input 20 names of the students of a class using array and print them in ascending order.

Q.10. (Total 8 Marks)

EITHER

a. Artificial Intelligence (AI) is the science of making intelligent machines, especially intelligent computer programs. It is related to the task of using computers to understand human intelligence; but AI does not have to confine itself to methods that are biologically observable.

Describe the use of artificial intelligence in defence industry and weather forecasting. Also write at least TWO benefits of using artificial intelligence in these fields.

OR

b. Describe a computer virus and also highlight at least THREE ways in which it affects data.

END OF PAPER

Please use this page for rough work

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